

Data Project Part 1 and 2

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```
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v forcats 1.0.0      v readr    2.1.5
## v ggplot2  3.5.1      v stringr  1.5.1
## v lubridate 1.9.3      v tibble   3.2.1
## v purrr     1.0.2      v tidyr    1.3.1

## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(readr)
library(ggplot2)
```

1.[2 marks] The first part of our PPDAC framework is to identify the problem you are addressing with these data. State the question you are trying to answer and let us know what type of question this is in terms of the PPDAC framework. A question statement should be as specific as possible. For example: Do students who regularly get 8 hours of sleep have fewer visits to the health center? This question is an example of an etiologic or causal question.

How does the age-adjusted death rate for heart disease vary across different states in the United States in 2017, and are there any observable patterns or trends based on geographic regions (e.g., Northeast, South, Midwest, West)?

2. [2 marks] Why is this question interesting or important? You could talk here about how existing data/studies suggest this might be important, how the findings might make an impact, how the findings might be used, or why you are personally interested in this question.

Heart disease is one of the leading causes of death in the United States, and understanding how its mortality rates vary across states can help identify regions that may need targeted public health interventions. If certain states or regions consistently show higher death rates, it could indicate disparities in healthcare access, lifestyle factors, or environmental conditions that need to be addressed. The findings from this analysis could inform policymakers and healthcare providers about where to allocate resources, such as funding for prevention programs, healthcare infrastructure, or public awareness campaigns. For example, if the South has higher heart disease death rates, it might prompt initiatives to improve diet, exercise, and smoking cessation programs in that region. Observing geographic patterns in heart disease mortality could reveal insights into the role of regional factors such as diet (e.g., prevalence of high-fat or high-sodium diets), socioeconomic status, or cultural practices, leading to more tailored and effective public health strategies.

3. [2 marks] What is the target population for your project? Why was this target chosen? (i.e., what was your rationale for wanting to answer this question in this specific population?)

The target population for this project is the general U.S. population across different states. This target population was chosen because understanding leading causes of death across the country can help inform public health policies, allocate healthcare resources effectively, and identify trends in mortality rates over time.

4. [2 marks] What is the sampling frame used to collect the data you are using? It may be helpful here to read any protocol papers, trial registration records, 'Readme' files or documentation that are associated with your dataset. If you have trouble identifying how the records/individuals were sampled, confirm with your supporting GSI that your dataset will be usable for the purposes of the class. Describe why you think this sampling strategy is appropriate for your question. To what group(s) would you feel comfortable generalizing the findings of your study and why?

The dataset categorizes deaths by state, cause of death, and year, meaning that it was collected by stratifying the population based on these factors. It contains representative samples that ensures coverage across different regions and reasons. This strategy ensures proportional representation of each state, making regional comparisons possible, comprehensively records major causes of death, and minimizes bias. Additionally, annual trend analysis is highly valuable for public health research. The dataset contains mortality data categorized by state, making it suitable for analyzing state-level trends.

5. [2 marks] Write a brief description (1-4 sentences) of the source and contents of your dataset. Provide a URL to the original data source if applicable. If not (e.g., the data came from your internship), provide 1-2 sentences saying where the data came from. If you completed a web form to access the data and selected a subset, describe these steps (including any options you selected) and the date you accessed the data.

The dataset is sourced from National Center for Health Statistics (NCHS) that documents NCHS Leading Causes of Death in the United States. It contains information on the mortality data for various causes of death across different US states and years, including variables such as number of deaths, age-adjusted death rates, and cause classifications. The data were collected to evaluate changes in mortality rates over time across different states and demographic groups. I found this dataset from a website (<https://catalog.data.gov/dataset/nchs-leading-causes-of-death-united-states/resource/9096aa3c-0d4b-42f1-bb01-284816d92a15>) and accessed the data on 02/21/2025.

6. [1 mark] Write code below to import your data into R. Assign your dataset to an object. Make sure to include and annotate this code in your submission (you can use a # to comment out regular text within code chunks to annotate).

```
data <- read_csv("NCHS_-_Leading_Causes_of_Death__United_States.csv",
  col_types = cols(
    Year = col_double(), # Numeric (Discrete)
    `113 Cause Name` = col_character(), # Categorical
    `Cause Name` = col_character(), # Categorical (Text)
    State = col_character(), # Categorical (Text)
    Deaths = col_double(), # Numeric (Discrete)
    `Age-adjusted Death Rate` = col_double() # Numeric (Continuous)
  ))
```

7. [3 marks] Write code in R (included in your submission with annotation) to answer the following questions: i) What are the dimensions of the dataset?

```
dimension=dim(data)
dimension
```

```
## [1] 10868      6
```


ii) What are the variable names of the variables in your dataset?

```
variable_name=colnames(data)
variable_name
```

```
## [1] "Year"                "113 Cause Name"
## [3] "Cause Name"          "State"
## [5] "Deaths"              "Age-adjusted Death Rate"
```

iii) Print the first six rows of the dataset.

```
first_six_row=head(data)
first_six_row
```

```
## # A tibble: 6 x 6
##   Year '113 Cause Name' 'Cause Name' State Deaths Age-adjusted Death R~1
##   <dbl> <chr>           <chr>      <chr>  <dbl>          <dbl>
## 1  2017 Accidents (unintention~ Unintention~ Unit~ 169936          49.4
## 2  2017 Accidents (unintention~ Unintention~ Alab~   2703          53.8
## 3  2017 Accidents (unintention~ Unintention~ Alas~    436          63.7
## 4  2017 Accidents (unintention~ Unintention~ Ariz~   4184          56.2
## 5  2017 Accidents (unintention~ Unintention~ Arka~   1625          51.8
## 6  2017 Accidents (unintention~ Unintention~ Cali~  13840          33.2
## # i abbreviated name: 1: 'Age-adjusted Death Rate'
```

8. [2 marks] Use the data to demonstrate a data visualization skill we have covered during Part I of the course. Choose a visualization relevant to your stated problem. Include your code in your submission. For example, you could visualize the distribution of our outcome with a histogram, or use a bar graph to represent the distribution of your exposure variable.

```
HD_2017 <- data %>% filter(`Cause Name` == "Heart disease", Year == 2017)
```

```
region_mapping <- c(
  'Connecticut' = 'Northeast', 'Maine' = 'Northeast', 'Massachusetts' = 'Northeast',
  'New Hampshire' = 'Northeast', 'Rhode Island' = 'Northeast', 'Vermont' = 'Northeast',
  'New Jersey' = 'Northeast', 'New York' = 'Northeast', 'Pennsylvania' = 'Northeast',
  'Illinois' = 'Midwest', 'Indiana' = 'Midwest', 'Michigan' = 'Midwest',
  'Ohio' = 'Midwest', 'Wisconsin' = 'Midwest', 'Iowa' = 'Midwest',
  'Kansas' = 'Midwest', 'Minnesota' = 'Midwest', 'Missouri' = 'Midwest',
  'Nebraska' = 'Midwest', 'North Dakota' = 'Midwest', 'South Dakota' = 'Midwest',
  'Delaware' = 'South', 'Florida' = 'South', 'Georgia' = 'South',
  'Maryland' = 'South', 'North Carolina' = 'South', 'South Carolina' = 'South',
  'Virginia' = 'South', 'West Virginia' = 'South', 'Alabama' = 'South',
  'Kentucky' = 'South', 'Mississippi' = 'South', 'Tennessee' = 'South',
  'Arkansas' = 'South', 'Louisiana' = 'South', 'Oklahoma' = 'South',
  'Texas' = 'South', 'Arizona' = 'West', 'Colorado' = 'West', 'Idaho' = 'West',
  'Montana' = 'West', 'Nevada' = 'West', 'New Mexico' = 'West', 'Utah' = 'West',
  'Wyoming' = 'West', 'Alaska' = 'West', 'California' = 'West', 'Hawaii' = 'West',
  'Washington' = 'West', 'Oregon' = 'West', 'District of Columbia' = 'South')
```

```
HD_2017 <- HD_2017 %>%
  mutate(Region = region_mapping[State])
```

```
HD_2017 <- HD_2017 %>% filter(Region != "NA")
HD_2017
```

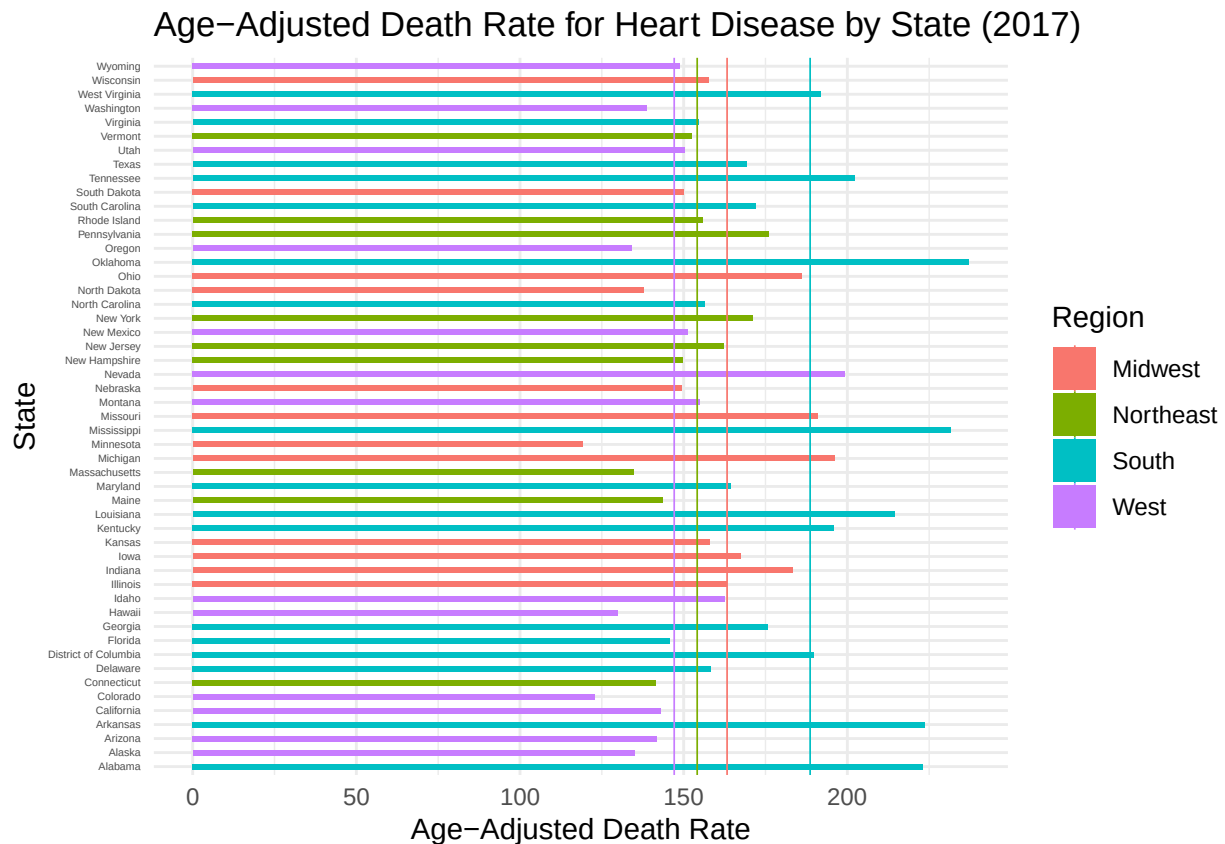
```
## # A tibble: 51 x 7
##   Year '113 Cause Name'      'Cause Name' State Deaths Age-adjusted Death R-1
##   <dbl> <chr>              <chr>      <chr>   <dbl>      <dbl>
## 1 2017 Diseases of heart (IO~ Heart disea~ Alab~   13110      223.
## 2 2017 Diseases of heart (IO~ Heart disea~ Alas~    814      135
## 3 2017 Diseases of heart (IO~ Heart disea~ Ariz~   12398     142.
## 4 2017 Diseases of heart (IO~ Heart disea~ Arka~    8270     224.
## 5 2017 Diseases of heart (IO~ Heart disea~ Cali~   62797     143.
## 6 2017 Diseases of heart (IO~ Heart disea~ Colo~    7060     123.
## 7 2017 Diseases of heart (IO~ Heart disea~ Conn~    7138     142.
## 8 2017 Diseases of heart (IO~ Heart disea~ Dela~    1990     158.
## 9 2017 Diseases of heart (IO~ Heart disea~ Dist~    1284     190.
## 10 2017 Diseases of heart (IO~ Heart disea~ Flor~   46440     146.
## # i 41 more rows
## # i abbreviated name: 1: 'Age-adjusted Death Rate'
## # i 1 more variable: Region <chr>
```

```
regional_avg <- HD_2017 %>% group_by(Region) %>% summarize(Average_Rate = mean(
  `Age-adjusted Death Rate`,
  na.rm = TRUE))
regional_avg
```

```
## # A tibble: 4 x 2
##   Region    Average_Rate
##   <chr>      <dbl>
## 1 Midwest    163.
## 2 Northeast  154.
## 3 South     189.
## 4 West      147.
```

```
plot_HD_2017 <- ggplot(HD_2017, aes(x = `Age-adjusted Death Rate`, y = State, fill = Region)) +
  geom_bar(stat = "identity", width = 0.4) +
  geom_vline(data = regional_avg, aes(xintercept = Average_Rate, color = Region), linewidth = 0.3) +
  labs(title = "Age-Adjusted Death Rate for Heart Disease by State (2017)",
       x = "Age-Adjusted Death Rate",
       y = "State") +
  theme_minimal() +
  theme(axis.text.y = element_text(size = 4))
```

plot_HD_2017



9. [2 marks] Describe the skill that you are demonstrating and interpret your findings. For example, if you have created a histogram, describe the central tendency, shape of the distribution, etc.

- Skill:

1. Data Filtering by using `filter()`: I used `filter(Region != "NA")` to ensure that only states with valid region classifications are included in the analysis. This step removes incomplete or irrelevant entries which may cause confusion in interpreting the plot. I also used `filter()` to filter the original dataset to include only rows where the cause of death is "Heart disease" and the year is 2017.
2. Data Aggregation by using `group_by()` and `summarize()`: I calculated the average age-adjusted death rate for each region with `group_by(Region)` and `summarize()` to get meaningful comparisons between the average regional death rate. By using `na.rm = TRUE`, this ensures that the averages are accurate by removing missing values.
3. Data Visualization by downloading `ggplot2` package: I used `ggplot` package generally for later plotting.
4. Horizontal Bar Chart by using `geom_bar()`: I showed each state's age-adjusted death rate with bars being colored by region for an easier comparison.
5. Regional Average Lines by using `geom_vline()`: I added vertical lines for each region's average death rate, allowing for immediate visual assessment of whether a state is above or below its regional average.
6. Minimal Theme by using `theme_minimal()`: I removed unnecessary grid lines and background elements, making the chart cleaner and easier for interpretation.
7. Data Categorization: I created a mapping of U.S. states to their corresponding geographic regions based on standard regional classifications (e.g., Northeast, Midwest, South, West).
8. Data Transformation by using `mutate()`: I added a new variable `Region` to the dataset by mapping the `State` column to its corresponding region from the `region_mapping` vector.

- Interpretation:

1. The Southern regions have higher age-adjusted death rates for heart disease because of the longer bars and higher regional average lines.
2. The Western regions generally exhibit lower death rates, mostly falling below their regional average.
3. States like California, Hawaii, and Colorado have significantly lower rates, suggesting potential regional health advantages while Mississippi, Oklahoma, and Alabama show several highest death rates, exceeding both their regional and likely national averages.
4. States above their regional averages like Mississippi in the South may benefit from targeted public health interventions, and States that fall below their regional average like California in the West may serve as case studies for developing better heart disease prevention strategies.

11.[3 marks] Describe the type of theoretical distribution that is relevant for your data. • What type of variable(s) are you investigating (continuous, categorical, ordinal, etc)?

The dataset consists of continuous, discrete, and categorical variables. Age-adjusted Death Rate is continuous variables. Discrete variables are Deaths and Year. Nominal categorical variables are State, Cause Name, and Region.

- What theoretical distribution that we have talked about would potentially be appropriate to use with these data (Normal, Binomial, Poisson. . .)

For Age-adjusted Death Rate, the Normal distribution is a appropriate choice since the death rates represent the averages over large populations, which tend to follow a bell-shaped (normal distribution) due to the Central Limit Theorem. And also the normal distribution models continuous variables. This distribution allows for comparisons between states and trend analysis over time. Furthermore, the Poisson distribution is appropriate for modeling the number of deaths, as it is commonly used for count data for events that occur independently over a fixed time or space. Because deaths are discrete occurrences, the Poisson model helps analyze variations in mortality patterns across states and causes.

- Why is this an appropriate model for the data you are studying? (HINT: what are the assumptions of this distribution?)

For age-adjusted death rates, the normal distribution is an appropriate model because death rates are continuous and usually summed over a large number of population data. The Central Limit Theorem suggests that when calculating mortality rates for many individuals, the distribution of these means tends to be normally distributed. To test this assumption, we can visualize the distribution of age-adjusted mortality rates using a histogram or Q-Q plot. If the distribution is symmetrically bell-shaped with no skewed tails, the normality assumption is valid. In addition, the Poisson distribution is suitable for modeling the number of deaths because it fits the count data for events that occur independently in a fixed period (year) and state. One assumption of the Poisson model is that the mean and variance are approximately equal. To test this assumption, we can compare the mean and variance of the number of deaths in different states and from different causes. If the variance greatly exceeds the mean, then a negative binomial model may be more appropriate.

12. [2 marks]What are the parameters that define this distribution? - Calculate these parameters for your data.

The normal distribution is defined by the mean and the standard deviation. In the context of our problem, the mean of the age-adjusted death rate is approximately 127.56, and the standard deviation is approximately 223.64.

13.[2 marks] Use your outcome data to calculate a probability. Provide an equation (use fpr,a; probability notation) that describes this probability. Note whether this probability is a conditional or a marginal probability. For example if my outcome variable is height in inches, I might calculate the probability that an individual in the dataset has a height of greater than 5 feet 10 inches. $P(\text{height} \geq 70 \text{ inches}) = ?$ This would be a marginal probability.

$P(\text{age-adjusted death rate} \geq 100) = 0.549$, or 54.9%

```
mean_rate <- mean(data$`Age-adjusted Death Rate`, na.rm = TRUE)
sd_rate <- sd(data$`Age-adjusted Death Rate`, na.rm = TRUE)
mean_rate
```

```
## [1] 127.5639
```

```
sd_rate
```

```
## [1] 223.6398
```

```
pnorm(100, mean=127.5639, sd=223.6398, lower.tail=FALSE)
```

```
## [1] 0.549046
```

14.[2 marks] What type of variable is your primary exposure of interest? If this variable is a demographic variable (age, gender identity, race/ethnic identity) explain how the categories of this variable are defined and what the rationale is for this (for example if gender identity is being used, is the idea to capture something about biology ie using gender identity as a marker for genetic or phenotypic sex, or as a marker of social exposures). If your data is not from a randomized trial where your exposure of interest was randomly assigned, are there important factors that may have affected how this exposure was distributed?

The primary exposure of interest in this dataset is state, which isn't a demographic variable such as age, gender identity, or race/ethnicity. Since the data doesn't come from a randomized trial, the exposure variable was not randomly assigned. People live in a specific state based on a variety of non-random factors like socioeconomic status, employment opportunities, and family ties. These underlying factors can affect the exposure distribution.

15. [4 marks] Use your data to create a visualization of your data that begins to explore your research question. Include code in R, a visual of some kind and text interpretation. For example, if your outcome is height of children at age 10 and your predictor variable is exposure to food insecurity in the first year of life, you could provide a histogram of height among children exposed to food insecurity and a separate histogram of height among children not exposed to food insecurity. Make sure you describe your interpretation of the results.

```
data <- read_csv("NCHS_-_Leading_Causes_of_Death__United_States.csv",
  col_types = cols(
    Year = col_double(), # Numeric (Discrete)
    `113 Cause Name` = col_character(), # Categorical
    `Cause Name` = col_character(), # Categorical (Text)
    State = col_character(), # Categorical (Text)
    Deaths = col_double(), # Numeric (Discrete)
    `Age-adjusted Death Rate` = col_double() # Numeric (Continuous)
  ))
```

```
HD_2017 <- data %>% filter(`Cause Name` == "Heart disease", Year == 2017)
```

```
region_mapping <- c(
  'Connecticut' = 'Northeast', 'Maine' = 'Northeast', 'Massachusetts' = 'Northeast',
  'New Hampshire' = 'Northeast', 'Rhode Island' = 'Northeast', 'Vermont' = 'Northeast',
  'New Jersey' = 'Northeast', 'New York' = 'Northeast', 'Pennsylvania' = 'Northeast',
  'Illinois' = 'Midwest', 'Indiana' = 'Midwest', 'Michigan' = 'Midwest',
  'Ohio' = 'Midwest', 'Wisconsin' = 'Midwest', 'Iowa' = 'Midwest',
  'Kansas' = 'Midwest', 'Minnesota' = 'Midwest', 'Missouri' = 'Midwest',
  'Nebraska' = 'Midwest', 'North Dakota' = 'Midwest', 'South Dakota' = 'Midwest',
  'Delaware' = 'South', 'Florida' = 'South', 'Georgia' = 'South',
  'Maryland' = 'South', 'North Carolina' = 'South', 'South Carolina' = 'South',
  'Virginia' = 'South', 'West Virginia' = 'South', 'Alabama' = 'South',
  'Kentucky' = 'South', 'Mississippi' = 'South', 'Tennessee' = 'South',
  'Arkansas' = 'South', 'Louisiana' = 'South', 'Oklahoma' = 'South',
  'Texas' = 'South', 'Arizona' = 'West', 'Colorado' = 'West', 'Idaho' = 'West',
  'Montana' = 'West', 'Nevada' = 'West', 'New Mexico' = 'West', 'Utah' = 'West',
  'Wyoming' = 'West', 'Alaska' = 'West', 'California' = 'West', 'Hawaii' = 'West',
  'Washington' = 'West', 'Oregon' = 'West', 'District of Columbia' = 'South')
```

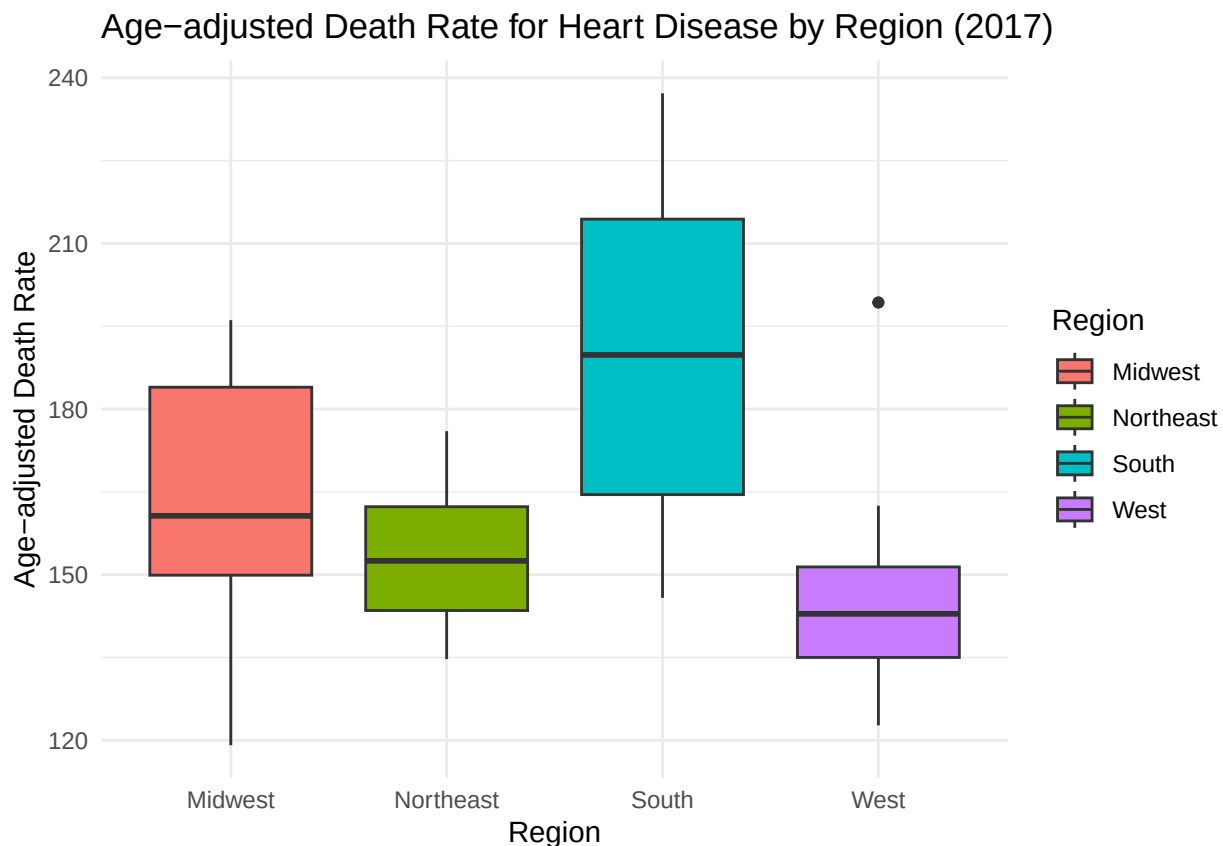
```
HD_2017 <- HD_2017 %>%
  mutate(Region = region_mapping[State])
```

```
HD_2017 <- HD_2017 %>% filter(Region != "NA")
HD_2017
```

```
## # A tibble: 51 x 7
##   Year '113 Cause Name'      'Cause Name' State Deaths Age-adjusted Death R-1
##   <dbl> <chr>              <chr>         <chr>   <dbl>          <dbl>
## 1 2017 Diseases of heart (IO~ Heart disea~ Alab~   13110          223.
## 2 2017 Diseases of heart (IO~ Heart disea~ Alas~    814          135
## 3 2017 Diseases of heart (IO~ Heart disea~ Ariz~  12398          142.
## 4 2017 Diseases of heart (IO~ Heart disea~ Arka~   8270          224.
## 5 2017 Diseases of heart (IO~ Heart disea~ Cali~  62797          143.
## 6 2017 Diseases of heart (IO~ Heart disea~ Colo~   7060          123.
```

```
## 7 2017 Diseases of heart (I0~ Heart disea~ Conn~ 7138 142.
## 8 2017 Diseases of heart (I0~ Heart disea~ Dela~ 1990 158.
## 9 2017 Diseases of heart (I0~ Heart disea~ Dist~ 1284 190.
## 10 2017 Diseases of heart (I0~ Heart disea~ Flor~ 46440 146.
## # i 41 more rows
## # i abbreviated name: 1: 'Age-adjusted Death Rate'
## # i 1 more variable: Region <chr>
```

```
plot_15 <- ggplot(HD_2017, aes(x = Region, y = `Age-adjusted Death Rate`, fill = Region)) + geom_boxplot()
plot_15
```



-Interpretation: I chose to use boxplots to show the distribution of age-adjusted death rate across 4 major regions in the United States. There are regional differences in cultural norms, socioeconomic status, healthcare access and medical infrastructure acting as risk factors of age-adjusted death. For example, the Southern parts have the highest median death rate and greater variability, showing a greater disparity within this region and more health burdens. And the Midwest shows relatively higher rates, while the Northeast and West have lower medians and more centralized distributions.